The Pan-Disciplinary Architecture of Human Song: Evolutionary Universals, Vocal Tract Bioacoustics, Striatal-Cortical Neurobiology, Neural Synthesis, and Algorithmic Compression

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Abstract

Human song represents a complex biological, acoustic, and cultural phenomenon operating at the intersection of evolutionary signaling, laryngeal biomechanics, striatal-cortical neuroplasticity, generative signal processing, and digital media economics. Cross-cultural ethnographic and discographic analyses across 315 human societies confirm that vocal music—both with and without words—is a pan-human universal. Rather than functioning as an arbitrary cultural artifact, song varies systematically along three latent dimensions of formality, arousal, and religiosity, exhibiting substantially greater variance within individual societies than between them. Acoustic forms reliably predict primary behavioral functions—specifically infant care, healing, dance, and love—allowing naive adult and pediatric listeners worldwide to infer behavioral context from spectro-temporal cues alone. Bioacoustically, the transition from propositional speech to song is governed by fundamental frequency (f_0) intra-syllabic stability, discrete pitch interval quantization, and isochronous inter-onset intervals. In unamplified classical vocal production, nonlinear source-filter coupling within the epilaryngeal tube clusters the third, fourth, and fifth formants (F_3, F_4, F_5) into the 2.4–3.4 kHz Singer's Formant prominence, while high-pitch soprano phonation relies on active osteo-muscular tuning of F_1 to f_0. Neurobiologically, song engages an anatomically dissociated mesolimbic dopaminergic circuit—recruiting the dorsal striatum (caudate nucleus) during temporal anticipation and the ventral striatum (nucleus accumbens) during peak affective consumption—while offering a structural bypass for post-stroke expressive aphasia via Melodic Intonation Therapy (MIT), which

drives white-matter remodeling in the homologous right arcuate fasciculus. Computationally, modern Singing Voice Synthesis (SVS) models approximate these physiological dynamics through shallow diffusion, multinomial-Gaussian pitch diffusion (P-DDPM), and flow-matching autoencoders coupled with differentiable digital signal processing (DDSP) harmonic-plus-noise vocoders. Simultaneously, the contemporary macro-structure of recorded song is undergoing rapid evolutionary compression driven by streaming platform monetization thresholds (the 30-second rule) and skip-rate recommendation algorithms, truncating introductory segments, eliminating traditional bridge sections, and contracting track durations across global genres.

Evolutionary Ethnomusicology and Cross-Cultural Universality

For over a century, anthropological and ethnomusicological discourse debated whether musical behavior shares universal structural regularities emerging from shared human cognitive architecture or consists entirely of culturally contingent idioms. Early comparative analyses of global recordings, such as opportunistic samples drawn from the *Garland Encyclopedia of World Music*, indicated that features including lexical words, chest voice phonation, and an isochronous beat appear in a majority of global songs. However, opportunistic sampling lacked formal controls for ethnographer bias, selective nonreporting, and geographic autocorrelation, and examined acoustic form in isolation from behavioral context.

To resolve these limitations with objective scientific standards, the Natural History of Song (NHS) project constructed two parallel cross-cultural corpora: the NHS Ethnography and the NHS Discography. The NHS Ethnography compiled digitized primary ethnographic descriptions from the Human Relations Area Files (eHRAF), examining 315 societies overall and focusing deeply on the Probability Sample File—a geographically stratified pseudorandom sample of 60 traditionally living societies drawn from each of Murdock's 60 cultural clusters across 30 distinct geographic regions, where paragraphs are indexed using more than 750 Outline of Cultural Materials (OCM) identifiers such as infant care, mourning, or warfare. Complementing the textual record, the NHS Discography assembled field recordings of vocal performances across 86 mostly small-scale societies covering 30 world regions, systematically sampling four core behavioral categories: dance songs, healing songs, love songs, and lullabies.

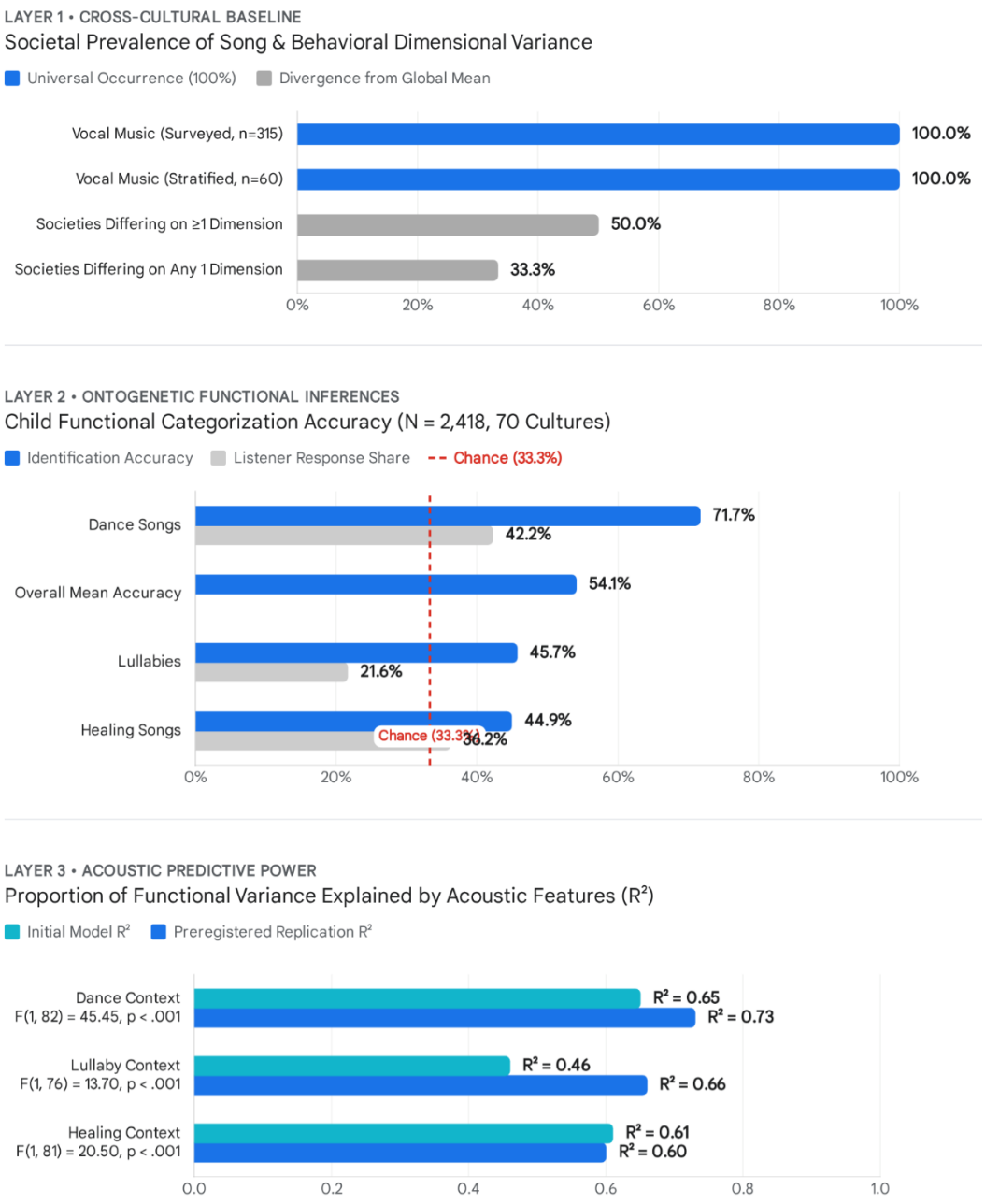


Figure 1: Objective Acoustic Features Drive Universal and Early-Developing Inferences of Song Function

Empirical metrics from the Natural History of Song corpora and developmental cross-cultural studies. **Top:** Prevalence of vocal music across 315 surveyed societies and 60 stratified cultural clusters, alongside the proportion of societies deviating significantly from global means across latent behavioral dimensions (formality, arousal, religiosity). **Middle:** Accuracy of 3-to-12-year-old children ($N = 2{,}418$) in identifying song contexts across 70 cultures relative to the 33.3% chance threshold, paired with listener response distributions. **Bottom:** Variance explained (R^2) in listener functional inferences by LASSO-selected acoustic features across exploratory and preregistered replication models.

Data sources: Science (Mehr et al., 2019), NHS Ethnography Corpus, Cognitive Science Proceedings, Journal of Experimental Psychology: General (Hilton et al., 2023)

Analysis of the ethnographic corpus establishes that music, explicitly including songs with words, is present in every single society examined. Dimensionality reduction of the high-dimensional ethnographic annotations—adjusted for selective nonreporting and ethnographer sampling bias—demonstrates that musical behaviors worldwide organize along three primary orthogonal dimensions: formality, arousal, and religiosity. Crucially, variance decomposition reveals that musical behavior varies substantially more within a single society across different behavioral contexts than it does across different societies performing the same behavioral function. At the same time, cultural diversity remains statistically robust, as approximately one-third (33.3%) of societies differ significantly from the global mean on any single dimension, and half (50.0%) of all societies deviate significantly from the global average on at least one dimension. Evaluation of 20 long-standing anthropological hypotheses confirms that song is universally and non-randomly associated with four foundational behavioral contexts: infant care (lullabies), healing ceremonies, dance, and love.

Evaluation of the NHS Discography via four complementary analytical representations—automated machine signal summaries, blind amateur listener ratings, expert musicological annotations, and manual melodic and rhythmic transcriptions—demonstrates that the acoustic features of a song systematically predict its primary behavioral function. Exploratory cross-cultural distributions further show that tonality is pervasive and potentially universal across human musical idioms, while rhythmic and melodic complexity follow systematic statistical distributions. Because musical form covaries with behavioral function across isolated small-scale societies, naive listeners can decode the communicative intent of unfamiliar vocal performances without cultural familiarity. When adult listeners across 60 countries heard field recordings from 86 societies, they reliably inferred whether songs were used for dancing, soothing an infant, or healing sickness.

Second-order developmental evidence indicates that these form-function mappings do not require extensive enculturation. In a large-scale cross-cultural investigation of children aged 3 to 12 years ($N = 2,418$) evaluating vocal excerpts from 70 world cultures, pediatric listeners identified the original behavioral context at rates significantly exceeding the 33.3% chance threshold both overall ($M = 54.1\%$, $SD = 21.6\%$, $p < .001$) and within each discrete category: dance songs (71.7%, $p < .001$), lullabies (45.7%, $p < .001$), and healing songs (44.9%, $p < .001$). While children exhibited an overall response bias—selecting "dance" most frequently (42.2%), followed by "healing" (36.2%) and "lullaby" (21.6%)—regression of song-wise guessing proportions against LASSO-selected objective acoustic features confirmed that children and adults rely on near-identical acoustic cues. Across both initial exploratory models and preregistered replications, objective acoustic parameters

accounted for 65% to 73% of variance in dance inferences ($F(1,84) = 52.66$, $R^2 = 0.65$; replication $F(1,82) = 45.45$, $R^2 = 0.73$), 46% to 66% of variance in lullaby inferences ($F(1,85) = 36.05$, $R^2 = 0.46$; replication $F(1,76) = 13.70$, $R^2 = 0.66$), and 60% to 61% of variance in healing inferences ($F(1,80) = 17.89$, $R^2 = 0.61$; replication $F(1,81) = 20.50$, $R^2 = 0.60$). Specifically, slower tempo and reduced rhythmic accentuation cued listeners that a performance functioned as a lullaby, whereas slower tempo, lower beat stability, greater melodic arpeggiation, unaccompanied solo texture, and major tonality signaled a healing context, generating a perceptual overlap where 40.20% of healing guesses were actual healing songs and 39.33% were lullabies. Cross-cohort comparisons between pediatric and adult models revealed no statistically significant divergence for lullabies ($\chi^2 = 6.45$, $p = 0.073$) or healing songs ($\chi^2 = 3.35$, $p = 0.217$), and only a minor divergence in dance sensitivity ($\chi^2 = 5.95$, $p = 0.005$). Furthermore, physiological monitoring demonstrates that human infants autonomously relax—exhibiting reduced arousal and heart rate—in response to foreign, unfamiliar lullabies presented a cappella, confirming an innate psychobiological sensitivity to maternal vocal soothing acoustics.

To enable deep phylogenetic modeling alongside broad typological comparison, the Expanded Natural History of Song Discography (2025) scaled this empirical infrastructure to $n = 1,007$ vocal excerpts spanning 413 languages across 79 language families and 10 rigorous behavioral contexts. By combining broad global sampling with dense sampling inside three major language families—Atlantic-Congo, Austronesian, and Indo-European—the expanded corpus allows computational disentanglement of vertical linguistic ancestry from horizontal geographic diffusion in shaping vocal diversity.

Table 1: Taxonomic Classification of Global Vocal Contexts in the Expanded NHS Corpus

Behavioral Context	Operational Inclusion Definition	Strict Exclusion Criteria	Statistical Discrimination (ppp-value)
Dance	Sung with the primary goal of a person or group dancing along to the vocal performance.	Songs accompanied by incidental dancing but performed for other primary goals.	$p < .005$ (permutation test vs. 1/k chance)

Behavioral Context	Operational Inclusion Definition	Strict Exclusion Criteria	Statistical Discrimination (ppp-value)
Healing	Sung within a formal healing ceremony with the explicit goal of curing physical or spiritual sickness.	Songs merely describing sick individuals or recounting a past epidemic.	$p < .005$ (permutation test vs. 1/k chance)
Love	Sung to express romantic or affectionate love directly to another person or describe currently felt love.	Songs concerning unrequited love, deceased loved ones, or affection for animals or property.	$p < .01$ (permutation test vs. 1/k chance)
Lullaby	Sung to an infant or young child with the goal of soothing, calming, or inducing sleep.	Songs designed to excite the child, play songs, or interactive singing games.	$p < .005$ (cross-cultural identification)
Play	Sung to excite an infant or child and actively engage them in play, including structured singing games.	Children's songs intended for calming, soothing, or sleep induction.	Classified within 10-context corpus ($n = 1,007$)
Procession	Sung to accompany a formalized march, entrance, or parade (e.g., weddings, funerals, leader introductions).	Processions characterized primarily by recreational or ceremonial dancing.	Classified within 10-context corpus ($n = 1,007$)

Behavioral Context	Operational Inclusion Definition	Strict Exclusion Criteria	Statistical Discrimination (ppp-value)
Mourning	Sung to express grief or sorrow regarding the death of a person in the present or past.	Songs performed for sick or dying individuals prior to death, or laments for non-mortal events.	Classified within 10-context corpus (n = 1,007)
Work	Sung to synchronize or accompany labor activities (planting, grinding, harvesting, tool-making).	Hunting songs performed prior to or celebrating successful hunts.	Classified within 10-context corpus (n = 1,007)
Story	Sung to recount historical or mythological events or narrate a sequence of actions by agents.	Lullabies or soothing songs that incidentally incorporate narrative stories.	Classified within 10-context corpus (n = 1,007)
Praise	Sung to express admiration for the traits or achievements of a person, animal, location, or object.	Songs expressing romantic love for a person or explicitly religious devotionals.	Classified within 10-context corpus (n = 1,007)

Bioacoustics of the Singing Voice and the Speech-to-Song Continuum

While spoken language and vocal song share the same peripheral anatomical apparatus—the pulmonary power source, the laryngeal vibrating vocal folds, and the supraglottal vocal tract filter—they occupy distinct regimes of spectro-temporal parameter space. Cross-cultural acoustic analyses confirm that speech and song are

differentiated globally by two primary physical markers: intra-syllabic pitch stability and temporal regularity of inter-onset intervals (IOI). In ordinary propositional speech, the fundamental frequency (f_0) glides continuously within and across syllables to encode rapid intonational prosody or lexical tone, exhibiting high intra-syllabic f_0 variability and high durational variability across syllabic units, with speech-to-song variability ratios reaching 5:1 in codified vocal comparisons. Conversely, song constrains f_0 into discrete, sustained plateaus within each vocalic nucleus—maximizing pitch stability—while lengthening vowel durations and organizing syllabic onsets around isochronous rhythmic grids, typically modeled at inter-onset intervals of 430ms to 600ms in sensorimotor synchronization paradigms.

The cognitive boundary between these two modes is probed directly by Diana Deutsch's speech-to-song illusion (2008, 2011). When a short spoken phrase containing relatively stable intra-syllabic f_0 contours is extracted and looped repeatedly without acoustic modification, listeners experience an involuntary perceptual phase shift in which the stimulus ceases to sound like speech and is perceived vividly as sung melody. Experimental manipulations demonstrate that this transformation relies on two strict acoustic conditions: first, the latent stability of the f_0 contour within individual syllables, which allows the auditory system to map continuous speech pitches onto discrete tonal intervals upon repetition; and second, the exact temporal preservation of syllable ordering and inter-onset timing. If intervening phrases are transposed slightly or temporal intervals are jittered, the perceptual shift collapses, proving that repetitive exposure recruits music-specific neural pitch-interval and rhythmic-entrainment templates that are suppressed during single-pass semantic decoding.

Table 2: Spectro-Temporal and Bioacoustic Comparison Across Vocal Production Modes

Acoustic / Physiological Parameter	Propositional Speech	Intermediate Mode (Sprechgesang / Chant)	Trained / Traditional Song
Intra-Syllabic f_0 Contour	Gliding, unstable; high intra-syllabic f_0 variability (up to 5:1 ratio vs. song).	Stylized prosodic inflection; replaces constant pitches with varying pitch contours.	Highly stable intra-syllabic f_0 plateaus; discrete interval steps.

Acoustic / Physiological Parameter	Propositional Speech	Intermediate Mode (Sprechgesang / Chant)	Trained / Traditional Song
Temporal / Rhythmic Organization	Irregular inter-onset intervals (IOI); high syllable duration variability.	Lengthened tempo; stylized into quarter- and eighth-note rhythmic grids.	Isochronous beat structure (typical IOI 430 - 600ms); sustained vowel nuclei.
High-Frequency Spectral Envelope (2.4–3.4 kHz)	Comparatively flat on Long-Term Average Spectrum (LTAS); separate F_3, F_4, F_5 peaks.	Moderate spectral energy; dependent on prosodic stress and vocal loudness.	Prominent **Singer's Formant Cluster** (F_3+F_4+F_5 convergence) exceeding -20dB vs. F_1.
Epilaryngeal Tube Configuration	Coupled continuously with main pharyngeal cavity; standard ventricular volume.	Variable laryngeal height; transitional aryepiglottic constriction.	Autonomous resonator near 3kHz via expanded laryngeal ventricles and narrowed epilarynx.
Formant Tuning (F_1 vs. f_0 Relationship)	Formants (F_1, F_2) vary solely to preserve phonetic vowel identity; $f_0 < F_1$.	Phonetic vowel targets maintained over exaggerated prosodic pitch excursions.	Active osteo-muscular tuning of $F_1 \approx f_0$ via jaw opening when $f_0 >$ normal F_1 (up to 880Hz).

At the physiological level, unamplified vocal projection in Western classical and operatic singing relies on specialized acoustic filtering within the vocal tract, first noted

by Bartholomew (1934) and Winckel (1952–1956) and systematically formalized by Johan Sundberg and Ingo Titze. When a trained singer projects over a symphony orchestra without electronic amplification, the voice does not simply increase subglottal vocal fold effort across all frequencies; rather, it reshapes the vocal tract filter to generate the Singer's Formant Cluster. On a Long-Term Average Spectrum (LTAS), the acoustic energy of a classical orchestra peaks around 450 – 500Hz and rolls off steeply above 1kHz, parallels that also characterize ordinary human speech murmur. The Singer's Formant creates a dense, broad spectral peak between 2.4kHz and 3.4kHz—centered around 2,840Hz in lower male voices such as baritones and basses, and approximately 3,000Hz in alto and contralto voices—occupying an open acoustic lane where orchestral masking is minimal and human auditory sensitivity is acute. Perceptually, this 3kHz energy concentration produces the ringing vocal timbre known as *squillo*.

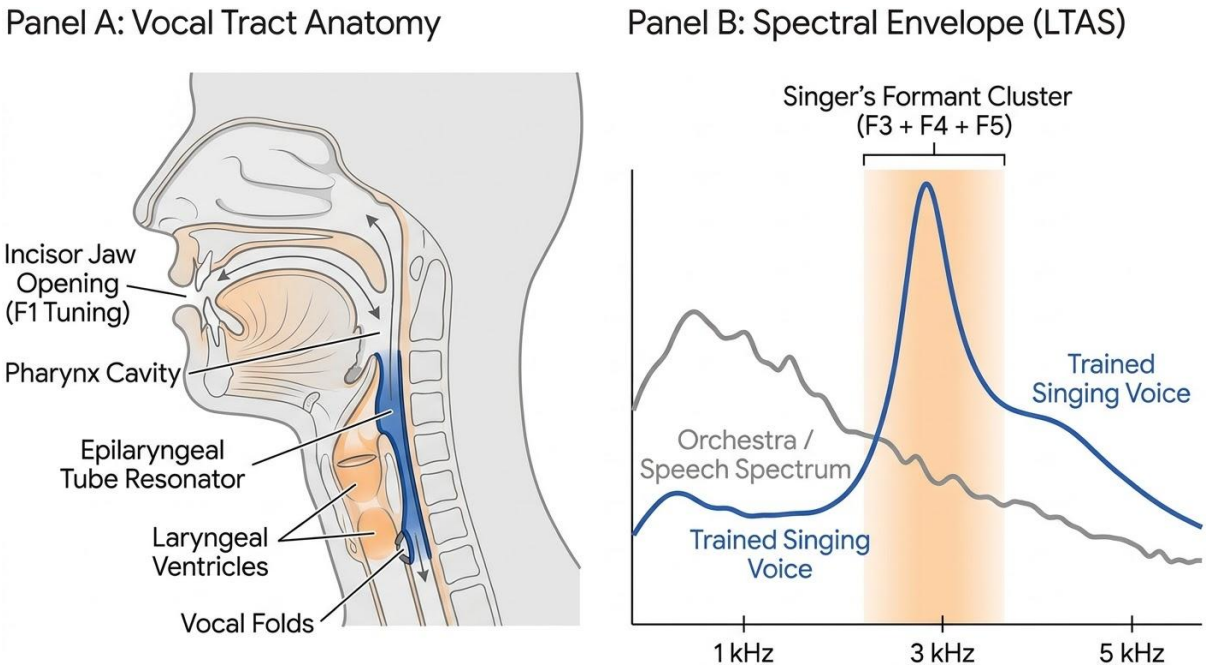


Figure 2: Epilaryngeal Resonance Clustering and Formant Tuning Generate the Singer's Formant Peak

Anatomical and spectral mechanics of classical vocal projection. Left: Sagittal cross-section of the vocal tract illustrating how enlarged laryngeal ventricles and a narrowed epilaryngeal tube create an autonomous resonator near 3 kHz (affiliated with F4), while variable jaw opening (measured at the incisors) tunes the first formant (F1) to match high fundamental frequencies. Right: Long-Term Average Spectrum (LTAS) comparison demonstrating how the clustering of the third, fourth, and fifth formants creates a prominent spectral peak between 2.4 and 3.4 kHz, allowing the unamplified singing voice to project above orchestral masking and ordinary speech.

Acoustically, the Singer's Formant is not an extra resonance added to the vocal tract, but the constructive convergence of the third, fourth, and fifth formants (F_3, F_4, F_5) into a single narrow frequency band. Sundberg's acoustic modeling demonstrated that in this

cluster, F_4 functions as a resonance specifically affiliated with the larynx tube—termed the epilaryngeal tube by Titze—which is the small supraglottal cavity bounded inferiorly by the vocal folds, anteriorly by the epiglottis, posteriorly by the arytenoids, and superiorly by the aryepiglottic folds. When the singer lowers the larynx, thereby expanding the laryngeal ventricles, and narrows the aryepiglottic opening so that the cross-sectional area of the epilaryngeal outlet is considerably smaller than that of the lower pharynx, the epilaryngeal tube decouples acoustically from the rest of the vocal tract. It then operates as an autonomous resonator tuned to a frequency directly between normal F_3 and F_4 (near 3kHz), drawing F_3, F_4 and F_5 together into mutual reinforcement. Clinically and acoustically, this prominence is quantified on an LTAS via the Singing Power Ratio (SPR) or by the criterion of Bloothooft and Plomp (1986), which defines a vowel as possessing a Singer's Formant when its 3kHz peak exceeds -20dB relative to the peak of the first formant (F_1).

While male and low female voices rely on the 3kHz Singer's Formant cluster, high soprano singing encounters a different physical boundary that necessitates formant tuning. In normal speech and low-pitched singing, the fundamental frequency f_0 lies well below the first formant F_1. However, when a soprano sings at high pitches (up to f_0 = 880Hz), the fundamental frequency rises far above the normal spoken F_1 of most vowels. Because vocal tract transfer functions attenuate acoustic energy when f_0 exceeds F_1, maintaining spoken articulatory positions at high pitch would waste glottal energy and yield a weak tone. Using artificial larynx vibrators held against the neck during silent articulation and incisor magnetometry—gluing micro-magnets to the upper and lower incisors of 10 singers performing two-octave scales across the vowels [u, o, α, a, e, i]—Sundberg and Skoog (1997) proved that sopranos systematically widen their jaw opening as pitch ascends once f_0 approaches F_1, actively tuning F_1 to track in close proximity to the first partial (f_0). At f_0 = 880Hz, partials are spaced so widely that each individual harmonic partial forms a discrete peak in the spectral envelope; by aligning F_1 with f_0, the singer achieves substantial acoustic amplification while dynamically tailoring subglottal pressure across ascending crescendos and harmonic transitions, even though extreme formant-matching modifies phonetic vowel quality across chromatic scales.

Neurobiology of Musical Reward and Clinical Neurorehabilitation

The universal endurance of song across human societies is anchored by its direct coupling to evolutionary reward circuitry and sensorimotor language networks in the human brain. Unlike primary biological rewards that directly support homeostatic survival, song is an abstract, temporally unfolding cognitive stimulus. Yet combined Positron Emission Tomography (PET) and functional Magnetic Resonance Imaging (fMRI) investigations by Salimpoor, Zatorre, and colleagues (2011, 2013) demonstrated that intense emotional responses to music trigger endogenous dopamine release within the mesolimbic striatum along two anatomically and temporally dissociated pathways.

During the anticipatory phase leading up to a peak musical moment—characterized by harmonic tension, rhythmic delay, or melodic expectation—dopamine is released predominantly in the dorsal striatum, specifically within the caudate nucleus. Because the caudate nucleus is densely interconnected with associative and sensorimotor cortices, this dorsal striatal activation encodes temporal prediction, structural expectation, and the cognitive anticipation of abstract acoustic resolution. Conversely, during the consummatory peak phase at the exact moment of affective resolution, dopaminergic transmission shifts anatomically to the ventral striatum within the nucleus accumbens (NAc). Subsequent fMRI valuation studies demonstrate that the perceived reward value of a musical stimulus is governed directly by the functional connectivity between the nucleus accumbens and the auditory cortex within the superior temporal gyrus. Thus, auditory cortical templates storing internalized tonal and rhythmic priors communicate continuously with the ventral striatum, translating acoustic patterns into hedonic neurochemical reinforcement.

Beyond hedonic processing in healthy populations, the lateralized neural architecture of song provides a vital therapeutic mechanism for neurological rehabilitation following cerebrovascular stroke. Post-stroke aphasia impairs language function in 21% to 40% of stroke survivors, generating 2.6 to 4.7 million new cases globally each year. In patients suffering from severe non-fluent (Broca's) aphasia, ischemic damage typically destroys cortical and deep subcortical tissue in the left inferior frontal gyrus (Broca's area) and severs the anterior segment of the left arcuate fasciculus (AF)—the primary C-shaped dorsal white-matter tract arching around the Sylvian fissure to map auditory phonological representations in the temporal lobe onto motor articulatory plans in the frontal lobe. While these patients exhibit profound deficits in spontaneous speech initiation, single-word repetition, and articulatory agility, clinicians long observed that they can frequently sing the lyrics of familiar and novel songs with remarkable fluency.

Capitalizing on this clinical dissociation, Albert, Sparks, and Helm (1973) developed Melodic Intonation Therapy (MIT), a hierarchically structured, intensive behavioral intervention designed to restore propositional verbal output by translating speech into sung prosody. Standard MIT operates across three progressive levels of difficulty—each comprising 20 functional everyday words or social phrases presented with visual cues—advancing the patient from unison intoning with a clinician, to delayed repetition, then to *Sprechgesang* (a facilitation technique halfway between singing and speech that retains the lengthened rhythmic cadence of song while replacing constant musical pitches with variable, exaggerated speech prosody), and finally back to autonomous spoken production. The therapeutic efficacy of MIT rests on two synergistic neuromodulatory components combined with high-intensity dosing, typically administered for 1.5hours per day across 4 to 5 days per week over 3 to 6 weeks, requiring a cumulative threshold exceeding 55hours for structural white-matter remodeling. The first component is stylized two-pitch melodic intonation, wherein spoken phrases are mapped onto two discrete musical pitches typically separated by a minor third or perfect fourth, assigning the high pitch to stressed syllables and the low

pitch to unstressed syllables, while lengthening tempo into simplified quarter- and eighth-note patterns of 4 to 8 notes. Because right-hemisphere auditory cortices are dominant for coding slower syllabic modulation patterns and global spectral pitch contours, slowing the syllable rate and stabilizing pitch shifts acoustic processing from damaged left-hemisphere rapid-temporal networks to intact right-hemisphere prosodic networks. The second unique component is synchronous left-hand tapping, in which the clinician guides the patient to tap their unimpaired left hand on each intoned syllable. This behavioral manipulation directly engages a right-hemispheric sensorimotor network in the right premotor cortex that coordinates both manual movements and orofacial articulatory movements, facilitating speech initiation via rhythmic anticipation, rhythmic entrainment, and auditory-motor coupling.

Longitudinal neuroimaging investigations have resolved long-standing debates regarding whether MIT promotes recovery via right-hemisphere compensation or left perilesional reactivation. Early Positron Emission Tomography (PET) studies using modified protocols such as French *Thérapie Mélodique et Rythmée* (TMR)—which trains intonation merely as an immediate facilitation cue rather than a hierarchical intensive program—reported post-treatment activation in left perilesional cortex. However, systematic reviews confirm that studies adhering strictly to the intensive original MIT protocol in chronic non-fluent patients consistently demonstrate right-hemispheric functional and structural reorganization. Specifically, paired pre- versus post-treatment Diffusion Tensor Imaging (DTI) tractography conducted by Schlaug and colleagues (2008, 2009, 2016) reveals a statistically significant hypertrophy—manifesting as increased fiber count, total tract volume, and fiber length—within the homologous right arcuate fasciculus (r-AF) and right inferior longitudinal fasciculus following more than 55hours of MIT. Consequently, a unified dual-route clinical consensus now governs post-stroke song rehabilitation: in patients with small focal lesions and low left-AF lesion load, intonation therapy facilitates left perilesional recovery; in patients with large left fronto-temporal infarctions where left language cortex is destroyed, intensive MIT drives structural remodeling of the right arcuate fasciculus to establish an alternative homolog pathway for expressive speech and feedback control, a process that can be further augmented by adjunctive repetitive transcranial magnetic stimulation (rTMS) or transcranial direct current stimulation (tDCS) targeting the right Broca's homolog.

Table 3: Clinical Candidacy, Structural Protocol, and Neural Mechanisms of Melodic Intonation Therapy (MIT)

Clinical / Neurobiological Dimension	Standard MIT Specification	Mechanistic Rationale & Empirical Evidence
Optimal Patient Candidacy	Severe post-stroke non-fluent (Broca's) aphasia; poor single-word repetition; effortful, slow speech initiation; intact auditory comprehension (Wernicke's area); emotionally stable.	Ensures intact auditory feedback decoding and preserved right-hemisphere substrates while targeting dorsal auditory-motor mapping deficits and verbal apraxia.
Neuroanatomical Lesion Profile	Unilateral left-hemisphere infarction encompassing or undercutting Broca's area with high anterior **left arcuate fasciculus (AF)** lesion load; no bilateral brain involvement.	Anterior left-AF damage predicts non-fluent speech severity; preserved right hemisphere is required to host homologous fronto-temporal compensatory remodeling.
Acoustic & Motor Intervention Components	(1) Two-pitch intoning (minor 3rd or 4th interval; high pitch = stressed syllables, low pitch = unstressed); (2) Quarter/eighth-note rhythmic lengthening; (3) Syllable-synchronous left-hand tapping.	Lengthened syllabic duration recruits right auditory cortex (dominant for slow syllable coding); left-hand tapping activates right premotor orofacial-manual coupling.
Hierarchical Progression & Dosing	3 levels (20 items/level): Unison intoning → delayed repetition → *Sprechgesang* (prosodic bridge) → normal	High-intensity, high-dose repetition prevents reliance on rote palliative phrases and

Clinical / Neurobiological Dimension	Standard MIT Specification	Mechanistic Rationale & Empirical Evidence
	speech; 1.5h/day over 3–6 weeks (>55h total for structural change).	drives activity-dependent myelination and axonal sprouting.
White-Matter & Cortical Plasticity (DTI / fMRI / PET)	Paired pre/post-DTI shows increased fiber count, tract volume, and length in the **right arcuate fasciculus (r-AF)** and right inferior longitudinal fasciculus (r-IFL).	Remodeled right AF facilitates auditory-to-motor mapping and feedback control in large LH lesions; smaller lesions utilize left perilesional pathways.

Computational Modeling and Neural Singing Voice Synthesis (SVS)

Translating the bioacoustic and prosodic complexities of human song into generative mathematical models represents one of the primary challenges in modern audio signal processing. While Text-to-Speech (TTS) synthesis requires generating intelligible spoken prosody from text tokens, Singing Voice Synthesis (SVS) must map symbolic musical scores—comprising lyric phonemes, discrete note pitches, and note durations—into high-resolution waveforms that strictly obey musical pitch contours (f_0), exact rhythmic timing, dynamic timbre variations, vibrato, and legato transitions. Furthermore, traditional SVS pipelines required labor-intensive fine-grained music scores created by extracting initial phoneme durations via the Montreal Forced Aligner and manually correcting phoneme boundaries, whereas contemporary architectures increasingly train on realistic music scores paired directly with professional vocal recordings.

The architectural evolution of neural SVS has progressed through four distinct generational paradigms. In the first paradigm, early neural SVS systems such as *ByteSing* and *HiFiSinger* adapted the non-autoregressive *FastSpeech* architecture, employing a phoneme encoder, a frame-level duration predictor to expand lyric features to mel-spectrogram frame lengths, and a transformer-based mel decoder paired with a Parallel WaveGAN neural vocoder. While computationally fast, standard transformer decoders optimized solely with L_1 or L_2 reconstruction losses suffer from spectral over-smoothing, failing to capture the highly dynamic spectral harmonics and micro-pitch modulations of expressive singing. To address cascading errors between acoustic

models and vocoders, a second paradigm emerged around end-to-end variational autoencoders such as *ViSinger* and *ViSinger2*, which integrated the *VITS* (Variational Inference with adversarial learning for end-to-end Text-to-Speech) framework and introduced explicit f_0 predictors to guide latent prior networks directly from score to waveform.

To overcome spectral over-smoothing in spectrogram generation, a third paradigm introduced denoising diffusion probabilistic models (DDPMs) via *DiffSinger* and *RMSSinger*. Rather than diffusing from pure Gaussian noise over hundreds of steps, *DiffSinger* introduced a shallow diffusion mechanism that generates a coarse mel-spectrogram via a FastSpeech-style decoder and intersects the forward diffusion trajectory at a shallow step, recovering rich high-frequency vocal harmonics with reduced inference latency. Extending diffusion directly to vocal prosody, *RMSSinger* (2023) observed that predicting continuous f_0 and discrete unvoiced (UV) flags via deterministic L_1 or L_2 regression flattens natural pitch expressiveness. *RMSSinger* therefore introduced the Pitch Diffusion Probabilistic Model (P-DDPM), which jointly models continuous f_0 (x_t) via Gaussian diffusion and categorical unvoiced states ($y_t \sim \{0, 1\}^K$) via multinomial diffusion across forward Markov perturbation chains:

$$q(x_t|x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{(1-\beta_t)}x_{t-1}, \beta_t I\right)$$

$$q(y_t|y_{t-1}) = \mathcal{C}(y_t|(1-\beta_t)y_{t-1} + \beta_t/K), \quad q(y_t|y_0) = \mathcal{C}(y_t|\overline{\alpha_t}y_0 + (1-\overline{\alpha_t})/K)$$

Where $\mathcal{C}$ denotes a categorical distribution parameterized by variance schedule β_t, and the reverse transition $p\left(y_{(t-1)}|y_t\right) = C\left(y_{(t-1)}\middle|\theta_{\text{post}}(y_t, \widehat{y_0})\right)$ approximates the posterior $q\left(y_{(t-1)}|y_t, y_0\right)$. During Stage 1 training, the acoustic model minimizes a composite objective combining Gaussian pitch diffusion ($\mathcal{L}_{\text{gdiff}}$), multinomial UV diffusion ($\mathcal{L}_{\text{mdiff}}$), word/phoneme duration loss ($\mathcal{L}_{\text{d}}$), and coarse mel-spectrogram distance $L_{(mel)} = \left|\left|mel_p - mel_g\right|\right|$:

$$L_1 = L_{(gdiff)} + L_{(mdiff)} + L_d + L_{(mel)}$$

In Stage 2, the primary network is frozen and a diffusion-based post-net is optimized via $\mathcal{L}_{\text{post}}$ conditioned on the coarse mel output mel_p to synthesize fine-grained formants and breath textures.

In the fourth and most recent paradigm, SVS architectures have converged on flow-matching autoencoders combined with Differentiable Digital Signal Processing (DDSP) harmonic-plus-noise synthesis to achieve state-of-the-art perceptual fidelity in as few as 1 to 8 inference steps. In this framework, an encoder transforms the raw 24 kHz audio waveform x—normalized to -26dB and segmented into 4-second word-aligned windows—into a time-aligned latent representation $z = E(x) \in R^{(d \times T)}$, where d is the feature dimension and T is the frame count. An Emformer-block decoder then decodes three explicit physical parameters: fundamental frequency F_0, spectral envelope S

(reconstructed via non-negative pseudo-inverse $S = M_0 10^{(S_c - \epsilon)}$ where $M_0 = \max(M^{-1}, 0)$ prevents negative reconstruction artifacts), and aperiodicity coefficient A. Using forward ($\mathcal{F}$) and inverse ($\mathcal{F}^{-1}$) Short-Time Fourier Transforms (STFT) with a 2048-sample window and 75% overlap, the DDSP vocoder generates the periodic vocal fold glottal source from a sum of sawtooth-like harmonic signals e_h and the turbulent breath component from Gaussian white noise $e_n(t) \sim \mathrm{N}(0,1)$:

$$x_h = \mathrm{F}^{-}1\big((1 - A)S\mathrm{F}(e_h)\big)$$

$$x_n = \mathrm{F}^{-}1\big(AS\mathrm{F}(e_n)\big)$$

$$\hat{x} = \lambda_h x_h + \lambda_n x_n$$

where λ_h and λ_n weight the harmonic and noise contributions, respectively. By embedding physical source-filter laws directly into the neural network graph and training with spectral reconstruction, vector quantization, and adversarial losses, flow-matching DDSP systems match or surpass heavy baselines like *ViSinger2* and *HiddenSinger* across both subjective Mean Opinion Score (MOS) evaluations—rated on a 5-point scale across 80 test samples—and objective acoustic metrics including Mel-Cepstral Distortion (MCD), F_0 Root Mean Square Error (F_0-RMSE), Voicing/Unvoicing F1 (V/UV\ F_1), and Real-Time Factor (RTF).

Table 4: Comparison of Neural Singing Voice Synthesis (SVS) Architectures and Evaluation Frameworks

Architecture / Model	Core Generative Mechanism	Pitch & Prosody Modeling	Vocoding / Waveform Synthesis	Primary Objective & Subjective Metrics
FastSpeech SVS (*HiFiSinger*, *ByteSing*)	Non-autoregressive feed-forward Transformer with frame duration expansion.	Deterministic L_1 / L_2 regression on continuous F_0 and discrete UV flags.	Parallel WaveGAN neural vocoder from mel-spectrograms.	Fast RTF, but exhibits spectral over-smoothing and flat vibrato.
VAE End-to-End	Variational autoencoder with	Internal F_0 predictor guiding	End-to-end latent-to-waveform	Strong baseline MOS and

Architecture / Model	**Core Generative Mechanism**	**Pitch & Prosody Modeling**	**Vocoding / Waveform Synthesis**	**Primary Objective & Subjective Metrics**
(*ViSinger*, *ViSinger2*)	adversarial learning (VITS backbone).	latent prior network alignment.	decoding without explicit mel cascading.	V/UV F_1; larger parameter footprint.
Shallow Diffusion (*DiffSinger*, *HiddenSinger*)	Denoising diffusion probabilistic model intersecting at a shallow step.	FastSpeech-based coarse prior with optional integrated F_0 predictor.	Neural vocoder conditioned on diffusion-refined mel-spectrograms.	High-detail formant texture; lower MCD than deterministic Transformers.
Pitch Diffusion (*RMSSinger*)	Two-stage coarse mel decoder + diffusion post-net $(L_1 + L_{(post)})$.	**P-DDPM**: Joint Gaussian diffusion for F_0 and multinomial diffusion for UV.	Diffusion post-net $(L_{(post)})$ + neural vocoder trained on realistic scores.	Superior F_0-RMSE and natural pitch dynamics on realistic scores.
Flow-Matching + DDSP (Emformer Autoencoder)	Conditional flow matching (1-step and 8-step inference) in latent space $z \in R^{(d \times T)}$.	Joint latent decoding of F_0, spectral envelope S, and aperiodicity A.	Differentiable harmonic-plus-noise STFT synthesis $(\hat{x} = \lambda_h x_h + \lambda_n x_n)$.	State-of-the-art MOS, RTF, MCD, F_0-RMSE, and V/UV F_1 with minimal parameters.

Macro-Structural Evolution of Song in the Algorithmic Streaming Era

While evolutionary biology and vocal physiology establish the invariant human capacity for song, the macro-structural architecture of commercial recorded song—its total duration, section ordering, introductory length, and harmonic modulation—is heavily governed by the technological medium of distribution. Historically, physical recording formats imposed hard mechanical boundaries on song runtime twice before the digital era. Between the 1930s and 1950s, the 78 rpm shellac disc enforced a strict physical ceiling of approximately 2minutes and 30seconds per side. From the 1960s through the early 2000s, the introduction of the long-playing (LP) vinyl record, magnetic cassette, and compact disc (CD) removed this physical constraint, establishing a multi-decade broadcast radio equilibrium. During the CD and early digital download era (1998–2018), hit recordings followed a standardized narrative architecture—Instrumental Intro, Verse 1, Pre-Chorus, Chorus, Verse 2, Pre-Chorus, Chorus, Bridge, Modulated Final Chorus, and Fade-Out Outro—resulting in a mean Billboard Hot 100 duration of 4:10 in 2000 (and 4:16 for UK number-one singles in 1998), gradually settling to a floor of 3:30 by 2018.

Since 2019, the dominance of on-demand streaming platforms such as Spotify and Apple Music alongside short-form video ecosystems has triggered a third historical phase: algorithmic re-compression. Unlike terrestrial radio, where listeners were passive and captive to programmed airplay, streaming interfaces grant instantaneous skip capability while enforcing two rigid economic and algorithmic mechanics. The first mechanic is the 30-second monetization threshold: on both Spotify and Apple Music, a playback session must reach 30 consecutive seconds before it registers as a counted stream and generates a contractual royalty payout. If a listener skips a track at 28 seconds, the artist and rights-holders receive zero revenue and zero stream attribution. According to Luminate's 2025 Mid-Year Music Report, 24.7% of all song plays across streaming platforms are abandoned prior to the 30-second mark, and for independent artists lacking established fanbase loyalty, Chartmetric's 2025 dataset reveals that pre-30-second abandonment climbs to 31% – 35%. Economically, reducing a track's pre-30-second skip rate from 35% to 15% across 10,000 listener impressions increases counted streams from 6,500 to 8,500, delivering a 30.7% increase in royalty revenue without additional marketing expenditure. The second mechanic involves algorithmic skip-rate penalties and hourly stream density: recommendation engines powering algorithmic playlists such as *Discover Weekly* and *Release Radar* weight early skip rate within the first 15 to 30 seconds and overall completion rate as primary quality signals during a track's initial 48 hours of release, such that high early skip rates immediately suppress algorithmic visibility. Simultaneously, because streaming payouts operate on a per-stream basis rather than duration-weighted listening time, total runtime functions as a distribution variable governed by the step function Streams/Hour = $\lfloor 3600 \div \text{Track Length in Seconds} \rfloor$. Consequently, a compressed 2:15 (135-second) streaming arrangement yields 26 potential stream events per hour of continuous playback, whereas

a 4:15 (255-second) CD-era arrangement yields only 14 stream events per hour in the same listening window.

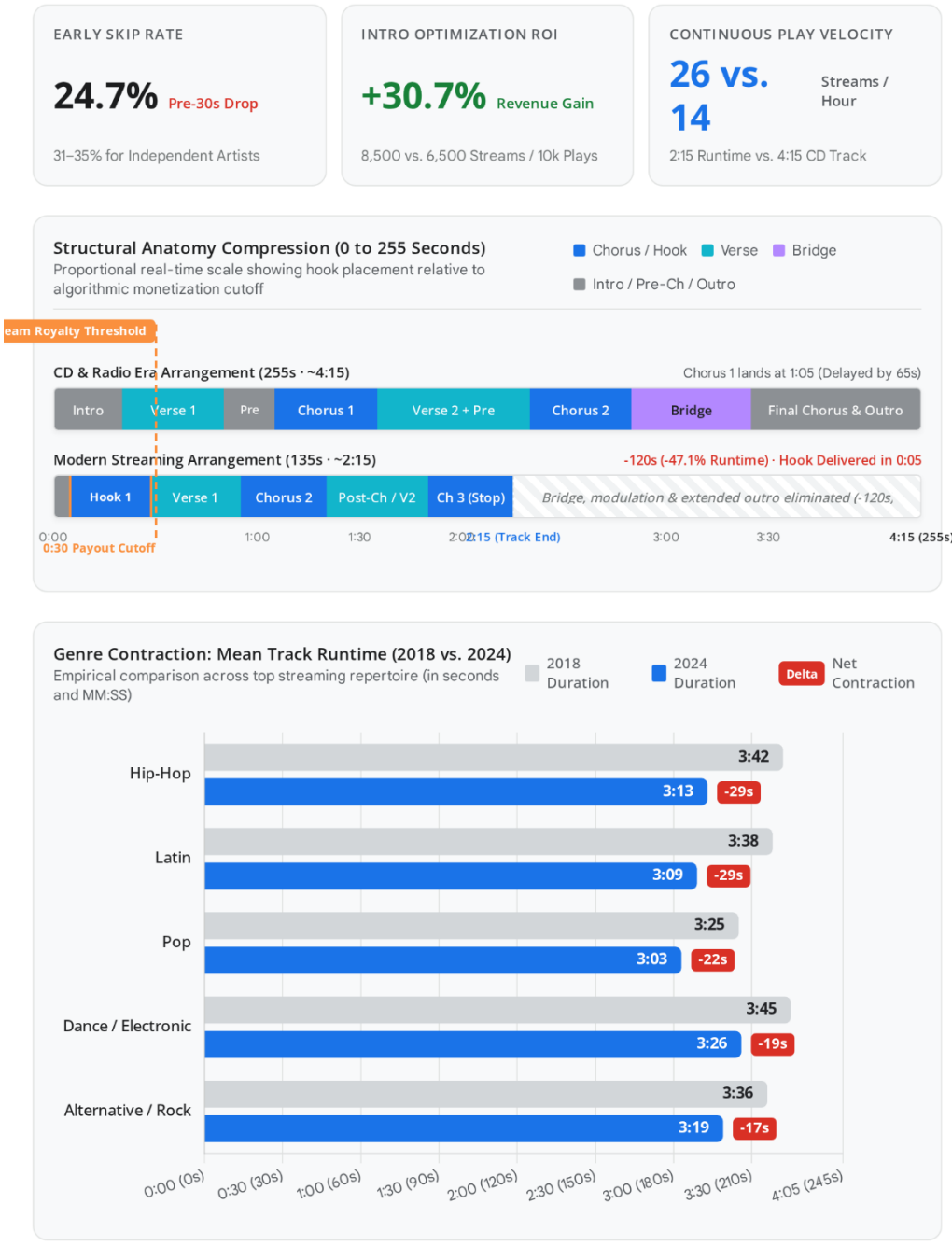


Figure 3: The 30-Second Monetization Rule Has Compressed Song Anatomy and Cut Track Durations Across All Major Genres

Economic drivers and structural manifestations of algorithmic song compression. **Top:** Streaming platform behavioral benchmarks from Luminate (2025) and Chartmetric (2025) alongside hourly playback capacity. **Middle:** Proportional structural comparison between a standard 4:15 (255s) CD/Download-era arrangement and a 2:15 (135s) Streaming-era arrangement, highlighting how modern tracks front-load the chorus

before the 30-second royalty payout threshold and eliminate the bridge and fade-out outro. **Bottom:** Mean track duration contraction across the five largest streaming genres between 2018 and 2024.

Data sources: Remasterify, International Journal for Multidisciplinary Research (IJFMR), MusicPulse & Luminate, TrustNode Logic & Chartmetric

Under these twin selection pressures, quantitative musicology identifies five systematic structural transformations across contemporary popular songwriting. First, song introductions have undergone severe truncation and sonic front-loading. In the CD and radio era, a 15-to-25-second instrumental intro established thematic atmosphere and provided broadcast DJs room to talk over the track before the vocal entered. Across hit songs between 1986 and 2015, average intro duration contracted from over 20seconds to approximately 5seconds, with contemporary hits frequently dropping a signature bassline, synth hook, or vocal within the first 2 to 5 seconds. A 2026 statistical analysis of the Billboard Year-End Hot 100 (N = 50 tracks)—defining intro duration strictly as the elapsed time from track onset to first verse vocal entry—found a mean intro length of 13.9seconds and a median of 12.0seconds, with 84% of charted hits introducing primary vocal content within the initial 20seconds. The underlying distribution exhibits noticeable positive skewness driven by isolated statistical outliers such as *"Redrum"*, *"Rich Baby Daddy"*, and *"Carnival"*, while a single-factor ANOVA confirms that genre acts as a statistically significant predictor of intro duration ($F(4,45) = 4.66$, $p = 0.003$, $\eta^2 = 0.293$), with Hip-Hop/Rap exhibiting the longest mean openings and Rock/Folk tracks demonstrating the shortest. Furthermore, widespread adoption of Digital Audio Workstations (DAWs) and click-track quantization has increased rhythmic consistency—measured via the tempo coefficient of variation $CV = \sigma / \mu$, where σ is the standard deviation of tempo measurements and μ is mean tempo—making it easier to tighten intros and introduce hooks earlier.

Second, pre-chorus sections have been compressed or rendered optional to accelerate hook delivery. Whereas CD-era arrangements delayed the first chorus until 45 to 75seconds into the recording, streaming arrangements deliver the primary chorus or vocal hook between 0 and 30seconds so that maximum melodic payoff occurs prior to the 30-second stream validation threshold. In addition, lyrical construction within the chorus has inverted: whereas older songwriting placed the primary "money line" at the end of the chorus, modern structure lands the primary hook in the very first line. Third, the traditional bridge section has been largely eliminated in favor of the post-chorus. Historically, the bridge appeared in 75% to 85% of charted songs, serving as the emotional heart and contrasting departure section. Because a slower departure section invites mid-track listener skips, bridge prevalence has collapsed to 15% – 25% of contemporary tracks, frequently replaced by a short, punchy post-chorus hook that lands immediately after the chorus to sustain listener reward without losing momentum. Fourth, uniform four-bar harmonic loops have largely displaced late-song key

modulations, which dropped from an occurrence rate of 15% – 20% in the CD era to below 3% – 5% in the streaming era. Fifth, extended fade-out outros have been replaced by abrupt hard stops or top-and-tail seamless loops, where the final bar's harmonic content connects directly back to the opening bar to prevent late-track skip penalties and encourage automatic replay.

Chartmetric's longitudinal tracking of the five largest commercial genres between 2018 and 2024 demonstrates that while runtime contraction is universal, its velocity varies by genre consumption mechanics. Hip-Hop (3:42 to 3:13) and Latin (3:38 to 3:09) experienced the steepest contraction, each averaging 29seconds shorter by 2024, followed by Pop (3:25 to 3:03, losing 22seconds), Dance / Electronic (3:45 to 3:26, losing 19seconds), and Alternative / Rock (3:36 to 3:19, losing 17seconds). Notably, Dance and Electronic music retains a higher duration floor (3:26) because club and DJ arrangement techniques require functional 16-bar and 32-bar phrase-mixing intros, breakdowns, and outros to manage tension, texture shifts, and beatmatched transitions across sets. Simultaneously, the decentralization of music production into consistent home studio environments—utilizing generative arrangement prototyping alongside DAWs to rapidly export 5-, 10-, 15-, 30-, and 60-second sync licensing cuts and instrumental stems—has institutionalized modular, compressed song architecture across the global recording industry.

Table 5: Structural Anatomy of Commercial Song Across Distribution Eras

Structural Parameter	78 rpm Shellac Era (1930s–1950s)	Broadcast Radio / CD Era (1960s–2018)	Algorithmic Streaming Era (2019–2026)
Primary Constraint Mechanism	Physical disc surface capacity (10-inch shellac groove limit).	Passive captive radio airplay; LP/CD unlocked physical ceiling.	**30-second stream validation rule**; early skip-rate penalties; pro-rata stream math.
Mean Track Duration	~ 2:30 maximum ceiling per side.	3:30 – 4:16 (4:10 US mean in 2000; 3:30 floor in 2018).	2:30 – 3:07 across pop charts (many hits running 2:00 – 2:30).
Hourly Playback	~ 24 plays per hour.	~ 14 stream events/hour (for	~ 26 stream events/hour (for a

Structural Parameter	78 rpm Shellac Era (1930s–1950s)	Broadcast Radio / CD Era (1960s–2018)	Algorithmic Streaming Era (2019–2026)
Yield (⌊3600/L⌋)		a 4:15 / 255s track).	2:15 / 135-second track).
Intro Duration (Time to Vocal)	Brief ensemble pickup (5 – 15 s).	15 – 25s instrumental ramp (>20s mean in 1986).	~ 5s pop hook drop; 13.9s Hot 100 mean 12.0s median; 84% < 20s).
Time to First Chorus	Integrated into AABA 32-bar form.	45 – 75seconds (after full Verse 1 + Pre-Chorus).	0 – 30seconds (front-loaded chorus or immediate hook tease).
Bridge Section Prevalence	Standard B-section release in AABA.	**75%–85%** of tracks feature a contrasting bridge.	**15%–25%** of tracks; largely omitted or replaced by a punchy **post-chorus**.
Key Modulation Rate	Rare outside theatrical arrangements.	**15%–20%** of tracks include a late-song key change.	**Below 3%–5%**; relies on continuous 4-bar harmonic loops.
Outro Design	Formal cadential resolution.	Extended instrumental or vocal **fade-out** (20 – 40s).	**Hard stop** or **top-and-tail seamless loop** to trigger automatic replay.

Synthesis and Conclusions

Synthesizing empirical findings across evolutionary anthropology, vocal tract physics, cognitive neuroscience, generative machine learning, and media economics reveals that human song operates as an integrated adaptive system wherein lower-level biological constraints govern higher-level cultural and technological expressions. At the ultimate evolutionary level, song is an invariant feature of human social organization across all surveyed societies, structured around low-dimensional behavioral axes of formality, arousal, and religiosity alongside acoustic form-function regularities that enable even young children to decode lullabies, healing chants, and dance rhythms without prior cultural exposure. At the proximate physiological level, the acoustic boundary separating song from speech—intra-syllabic f_0 stability, discrete pitch intervals, and isochronous inter-onset timing—is precisely structured to engage specialized right-hemispheric temporal-frontal networks and striatal dopaminergic reward circuits.

Crucially, cross-domain analysis exposes deep structural convergences across these fields. First, the bioacoustic decouplings that enable physical vocal projection—such as the autonomous epilaryngeal tube resonance generating the 3kHz Singer's Formant Cluster and osteo-muscular $F_1 - f_0$ formant tuning—are now explicitly mirrored in computational Singing Voice Synthesis (SVS), where purely black-box neural decoders have been superseded by Differentiable Digital Signal Processing (DDSP) and pitch-diffusion architectures (P-DDPM) that separate harmonic glottal excitation from aperiodic noise and spectral envelope filtering. Second, the neurobiological dissociation between caudate-mediated temporal anticipation and nucleus accumbens-mediated peak reward explains why modern streaming algorithms have reshaped commercial song structure so aggressively: by compressing song intros to a median of 12.0seconds, eliminating low-arousal bridges, and front-loading choruses ahead of the 30-second royalty threshold, contemporary production maximizes rapid ventral striatal dopaminergic delivery to prevent early listener abandonment, trading away the long-horizon dorsal striatal anticipation cycles that defined twentieth-century musical form. Understanding song across these five scales provides a unified quantitative framework for comparative human cognitive science, clinical aphasia neurorehabilitation, neural audio engineering, and cultural evolution.

Declarations

Author Contributions (CRediT Taxonomy)

SAMUELSON G: Conceptualization, Methodology, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization, Supervision, and Project Administration. **SAMUELSON G** solely conceived the pan-disciplinary theoretical framework, synthesized the cross-cultural ethnographic, bioacoustic, neuroimaging, generative neural synthesis, and commercial streaming

datasets, designed the structural tables and figures, and wrote and approved the final manuscript for publication.

Data and Corpus Availability

All empirical cross-cultural, acoustic, neurobiological, computational, and music-industry data synthesized in this study are derived from publicly archived scientific repositories and peer-reviewed corpora. Cross-cultural ethnographic and audio data are accessible via the Natural History of Song (NHS) Ethnography and Discography repositories, the Human Relations Area Files (eHRAF) Probability Sample File, and the open-access Expanded Natural History of Song Discography (n = 1,007 vocal excerpts across 413 languages) published under a Creative Commons license. Realistic singing score benchmarks and audio demonstrations for pitch diffusion modeling are archived publicly via the RMSSinger repository, and commercial streaming telemetry reflects published datasets from Luminate, Chartmetric, and Billboard Year-End Hot 100 statistical archives.

Ethical Compliance and Human Subjects Standards

All human behavioral, developmental, and clinical neuroimaging data analyzed within this synthesis—including cross-cultural pediatric listening studies (N = 2,418), infant physiological relaxation trials, Positron Emission Tomography (PET) and functional Magnetic Resonance Imaging (fMRI) striatal dopamine investigations, Diffusion Tensor Imaging (DTI) aphasia recovery protocols, and subjective Mean Opinion Score (MOS) listener evaluations—adhered to institutional ethical review standards and informed consent protocols established by their respective primary research institutions.

Competing Interests

The author (**SAMUELSON G**) declares no financial, commercial, or non-financial competing interests related to commercial streaming platforms, clinical neurorehabilitation protocols, or proprietary voice synthesis technologies.

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